MASTER'S THESIS DEFENSE · BIOMEDICAL ENGINEERING

Development of a Clinical Liver MRI Dataset for Liver Tumor Detection Using Deep Learning

A phase-aware, provenance-documented dataset and a controlled arterial-only versus four-phase nnU-Net experiment

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OUTLINE

What this presentation covers

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Background and problem

Liver cancer burden, multiphase MRI and the data bottleneck

02

Gap and research questions

What is missing in public liver MRI data, and four questions

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Methodology

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Contributions, limitations and future work

BACKGROUND

Liver cancer: a major global burden

3rd

leading cause of cancer-related death worldwide; among the six most diagnosed cancers [1]–[3]

~80%

of primary liver malignancies are hepatocellular carcinoma (HCC) [4]

~18%

overall five-year survival, driven mainly by stage at presentation [5]

Most patients present at intermediate or advanced stages, when resection, transplantation or ablation is no longer possible — so **early, accurate detection decides access to curative therapy** [1], [2], [6].

BACKGROUND

Multiphase MRI: rich information, hard to use

Why MRI

- Reference non-invasive modality for focal liver lesions, superior soft-tissue contrast, no ionizing radiation [3], [7]–[9]
- Dynamic contrast phases — native, arterial, portal venous, delayed (and hepatobiliary) — each show a different aspect of perfusion [2], [6], [10]
- Diagnosis depends on reading the full series together, not one image [10], [11]

Why it is hard

- No fixed intensity scale: gray values change with vendor, field strength and sequence [7], [8], [15]
- Breath-holds misalign the phases [5]–[7], [9]
- Manual 3D delineation takes hours per patient [12], [13]
- Expert lesion Dice ≈ 0.74–0.78 versus > 0.95 for whole liver [13], [15]

PROBLEM STATEMENT

Four challenges in turning clinical MRI into a detector

Multiphase DICOM integration

Combine several 3D contrast phases without losing perfusion information or distorting space [3], [7], [8]

Anatomical and boundary variability

HCC varies in size, shape and signal, often inside cirrhotic or fibrotic livers [3], [6], [15]

Spatial class imbalance

The tumor fills a tiny fraction of the volume, biasing losses toward background [3], [12], [21]

Small-lesion detection

Small, early lesions matter most but have the fewest voxels and weakest contrast [12], [15]

Premise of the thesis: reliable automated detection is today **more a data-engineering problem than an architecture problem** [3], [8], [15].

RESEARCH GAP

No standardized, leakage-free liver MRI dataset exists

CT

Public benchmarks are CT-dominated (LiTS); MRI collections are small, single-centre and rarely publish splits or provenance [7], [8], [17], [22]

29%

average VIDS compliance of public datasets; provenance only 8%, quality documentation 25% [22]

0.67 → 0.21

foreground Dice after enforcing patient-disjoint splits — slice-mixed splits inflate results [24]

Missing: a **phase-aware, provenance-documented clinical liver MRI dataset** that is machine-learning-ready and evaluated without leakage [7], [8], [22], [23].

Four questions, five objectives

RQ1 Can a clinical single-centre multiphase corpus be standardized into a traceable, phase-aware, ML-ready dataset?

RQ2 How do lesion size, class balance, visibility and annotation coverage constrain what a detector can learn?

RQ3 Does a four-phase input improve HCC detection and segmentation over arterial-only, under an identical protocol?

RQ4 How uncertain is the multiphase advantage, and what sample size would establish it?

Objectives

1. Acquisition and curation requirements [6], [7]
2. Reproducible annotation and provenance protocol [6], [13], [22]
3. Preprocessing and input-representation pipeline [7], [15], [25]
4. nnU-Net 2D benchmark: arterial-only vs four-phase [3], [7], [12]
5. Evaluation and reporting standards [22]–[24]

LITERATURE REVIEW

Liver is nearly solved; tumors are not

Model (family)	Data	Liver Dice	Tumor Dice	Ref
Attention U-Net++	ATLAS, 60 cases	0.94	0.85	[36]
3D U-Net + ensemble	Late-phase MRI, 107 patients	—	0.74	[15]
nnU-Net vs Swin UNETR	Multiphase T1-VIBE, 458 scans	0.97 / 0.95	0.56 / 0.42	[7]
nnU-Net (hepatobiliary)	HBP MRI, surgical planning	0.97	0.77	[12]
SWTR-Unet (CNN + transformer)	1.5 T MRI, 48 cases	0.98	0.81	[3]
Cross-modality 3D U-Net	CirrMRI600+	0.94	—	[32]

Liver Dice clusters at **0.93–0.98**; tumor Dice spans **0.44–0.85** with phase, dataset and label definition. Human inter-rater tumor Dice is only **0.78** [15] — part of the “error” is label noise.

LITERATURE REVIEW

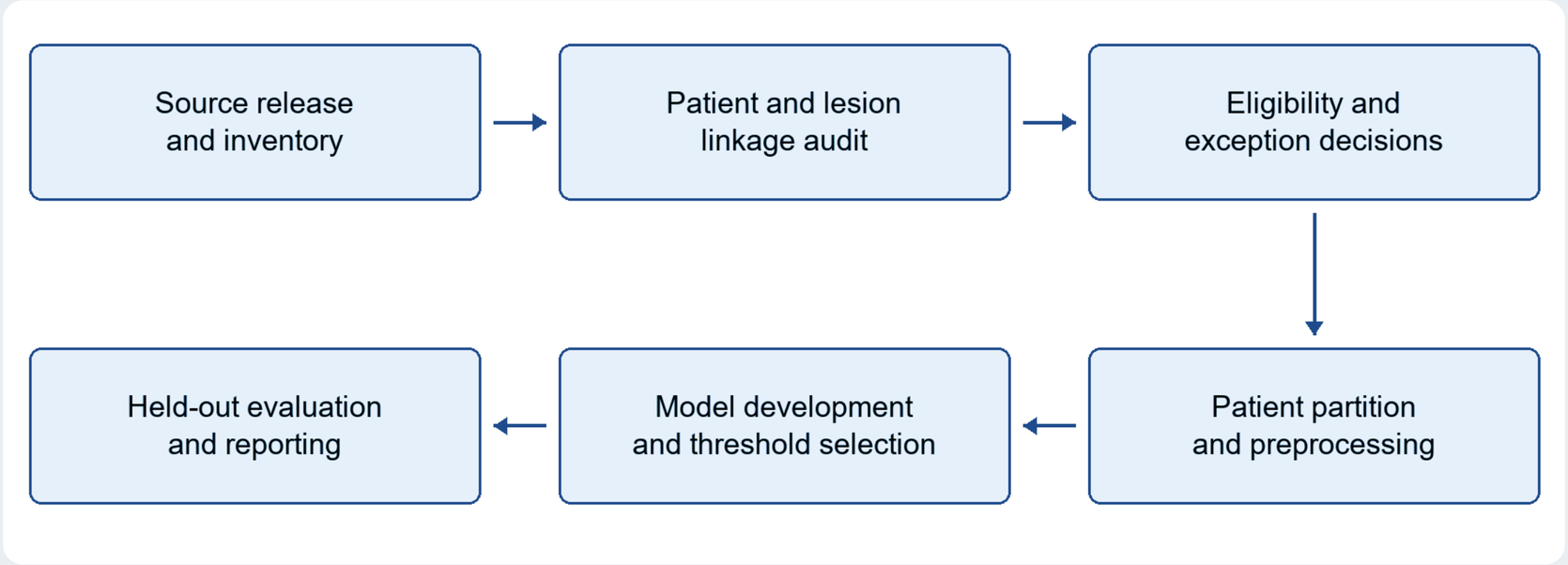
Public liver MRI datasets remain few and narrow

Dataset	Patients / scans	Sequences / phases	Annotations	Ref
OpenSwissHCC / ATLAS	60 (+30 private)	CE T1, 3 phases	Liver + HCC	[36], [37]
LiverHccSeg	17	T1 arterial	2-rater liver + HCC	[13], [38]
Duke Liver	105 / 2,146 series	Multi-sequence	Whole-liver masks	[8], [39]
CirrMRI600+	339 / 628 scans	T1 venous + T2	Cirrhotic liver	[8]
LiQA	440–610	T1/T2/DWI + 4 phases	Liver + fibrosis stage	[9]
LiTS (CT reference)	201 CT scans	CE-CT, 1 phase	Liver + tumor	[17]

Reported performance is partly a property of the dataset. This thesis adopts published conventions: **BIDS-like folders, NIfTI + JSON sidecars, per-annotation provenance and a split manifest**.

METHODOLOGY

A design-based, fully reproducible pipeline



Seven ordered stages — **audit, prepare, freeze, develop, calibrate, finalize, evaluate** — each writes a frozen, hashed artifact consumed by the next. Only the input representation differs between experiments.

METHODOLOGY

Cohort construction: 44 → 38 → 30 / 8 patients

44

Source subjects, all four dynamic phases; all pass the arterial-reference check

38

Retained after technical QC: 33 HCC-positive, 5 negative

30 development patients

26 positive / 4 negative · 38 lesions

8 held-out test patients

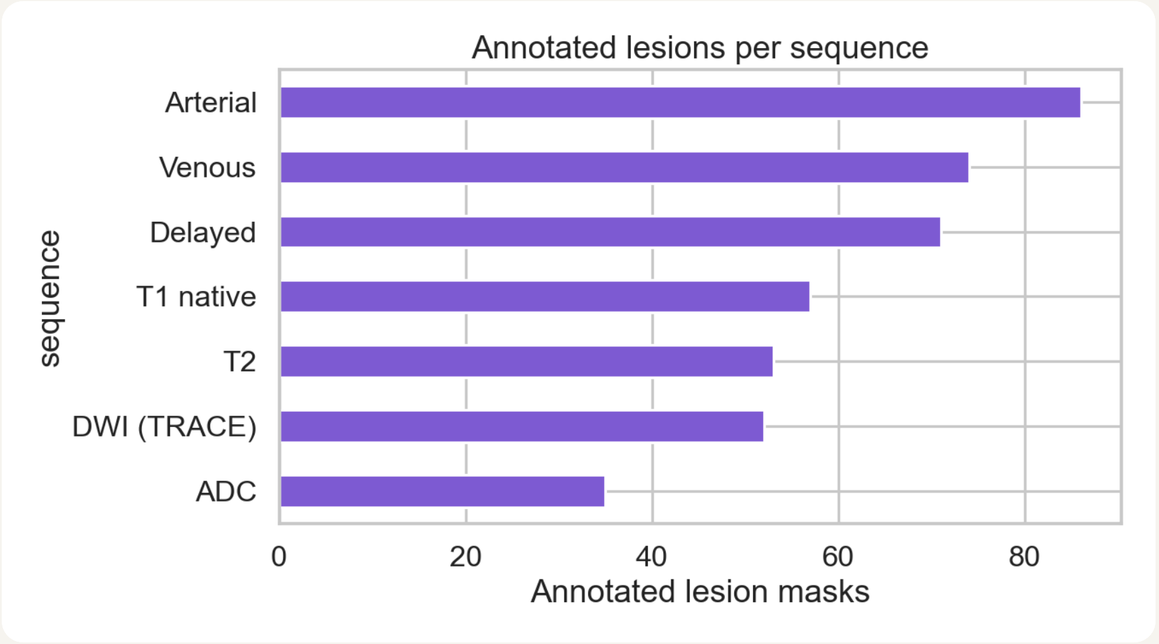
7 positive / 1 negative · 15 lesions

Source: Lausanne University Hospital, retrospective, 2011–2020; Skyra scanner for 72 of 89 lesion rows.

6 exclusions (one objective rule): the lesion mask was drawn on a different arterial acquisition than the highest-TTC one selected.

RESULTS · DATASET

A small-lesion, arterial-visible, mixed-truth dataset



89
annotated lesions in 39 positive subjects

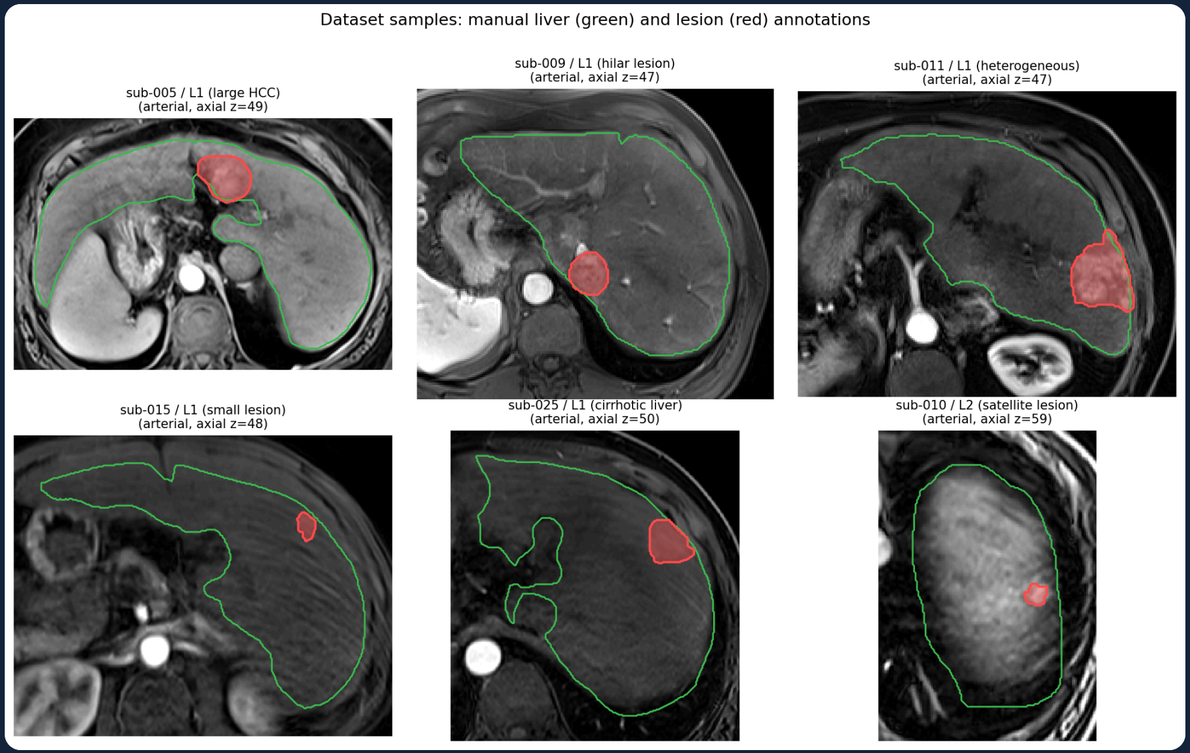
55/89
lesions under 20 mm (median 17 mm)

96.6%
visible on the arterial phase (ADC 36.0%)

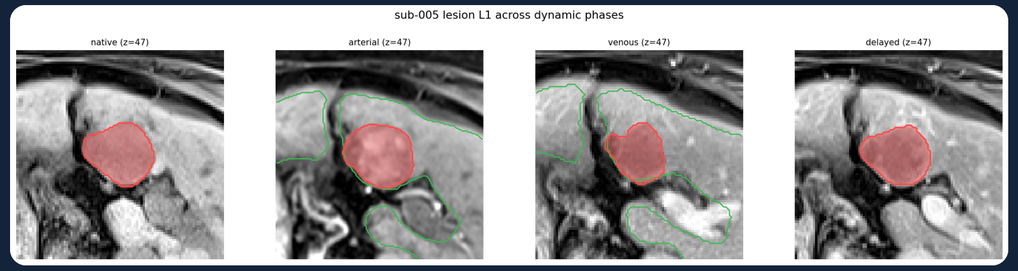
24/89
biopsy-proven; 38 LI-RADS 5, 27 unrecorded

Release: 766 NIfTI volumes, 427 manual lesion masks, 91 manual and 236 automated liver masks, 516 registration transforms; 43 of 47 metadata columns complete.

What the annotated data look like



Six lesions: liver contour (green), lesion mask (red)



One lesion across native, arterial, venous and delayed phases — the four-channel input of Experiment B

Lesions vary in size, location and contrast; each phase shows the same lesion differently, which is the information a multiphase model can exploit.

METHODOLOGY

One variable changes: the input phases

Experiment A — arterial only

1 channel: the arterial reference volume

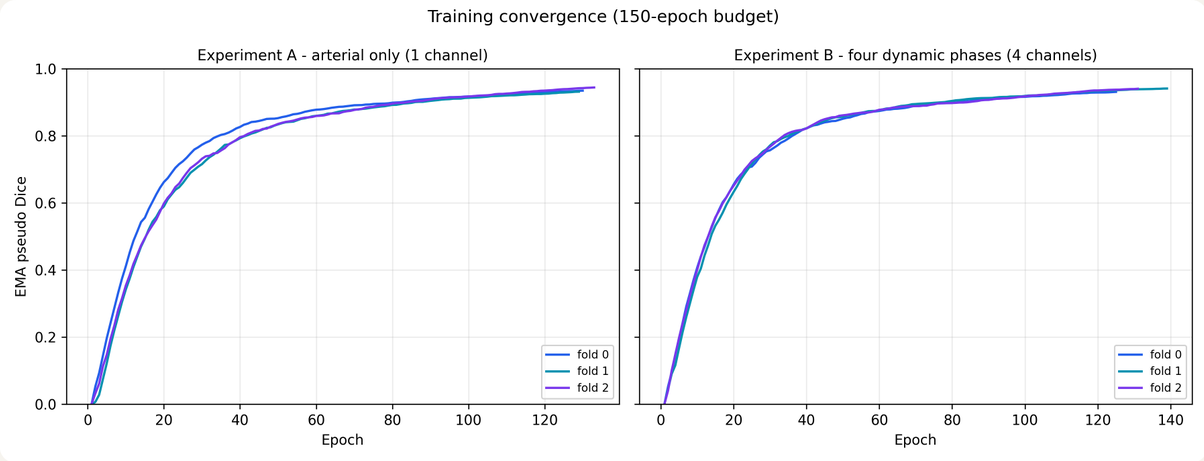
Experiment B — four phases

4 channels: native, arterial, venous, delayed

Held fixed in both arms	Setting
Preprocessing	Per-case foreground z-score; rigid registration (coverage ≥ 0.880)
Splits	Patient-disjoint 30 / 8; 3 class-stratified folds; seed 2026
Model and training	nnU-Net v2.6.2, 2D; 150 epochs; SGD, Dice + CE loss; EMA weights
Operating point	Max OOF lesion sensitivity at ≤ 1 FP per patient; frozen before test
Evaluation	Cardinality-first matching at IoU ≥ 0.10; 2,000 patient bootstraps

RESULTS · DEVELOPMENT

Training converged; out-of-fold, B finds twice as many lesions



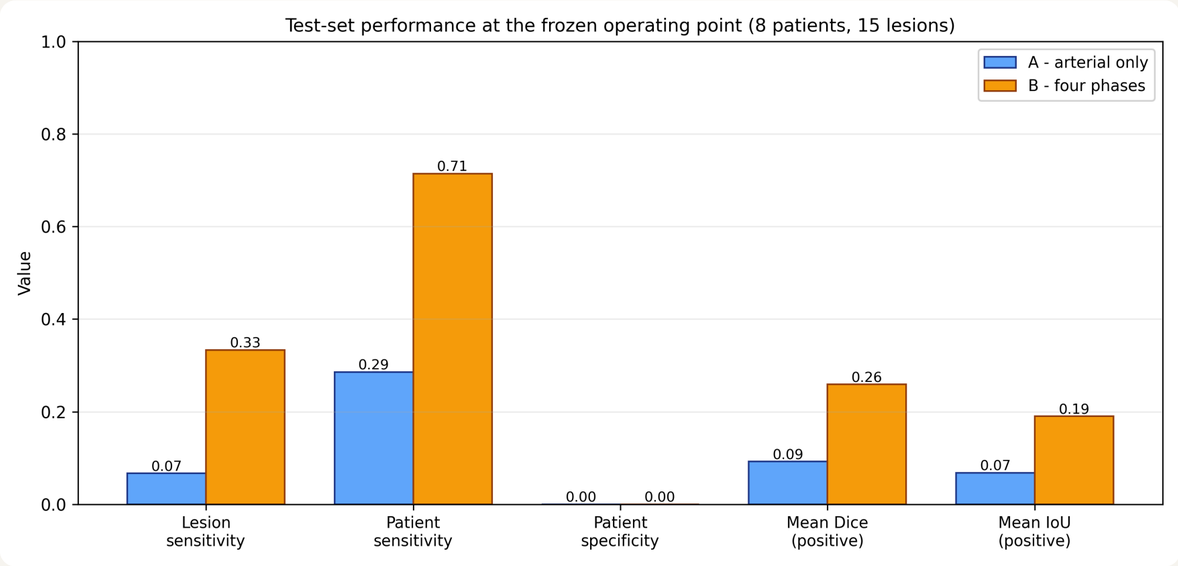
Final EMA pseudo-Dice 0.930–0.950 in all six jobs; curves flatten after epoch ~100–120.

Out-of-fold (30 patients)	A	B
Threshold	0.83	0.89
Matched lesions	6 / 38	13 / 38
Lesion sensitivity	0.158	0.342
FP per patient	0.567	0.733
Patient sensitivity	0.500	0.692
Mean Dice (positive)	0.100	0.220
ROC AUC	0.654	0.538

Experiment B was selected by the pre-set rule: it detected more than twice as many lesions under the false-positive budget.

RESULTS · HELD-OUT TEST SET (8 PATIENTS, 15 LESIONS)

Four phases detect five times more lesions



Lesions found

1 → 5

of 15 (A → B)

Mean Dice

0.092 → 0.259

2.8× higher

Patient sensitivity

0.29 → 0.71

PPV 0.67 → 0.83

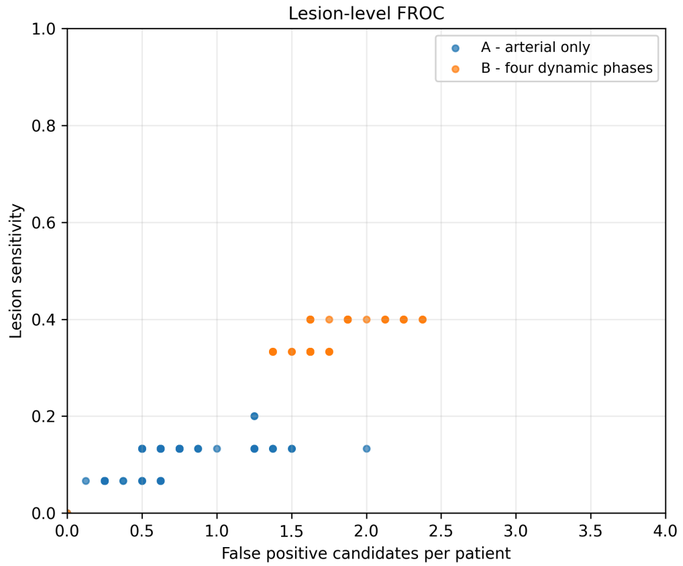
FP per patient

0.25 → 1.50

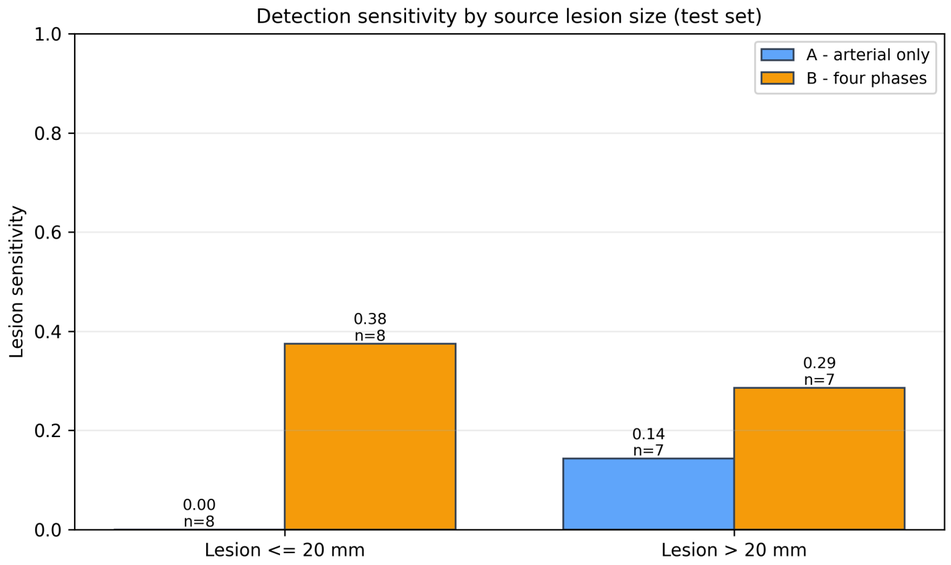
the price of sensitivity

Specificity, NPV and ROC AUC rest on a single negative test patient (n = 1) and are not interpretable.

The gain holds across operating points and sits in small lesions



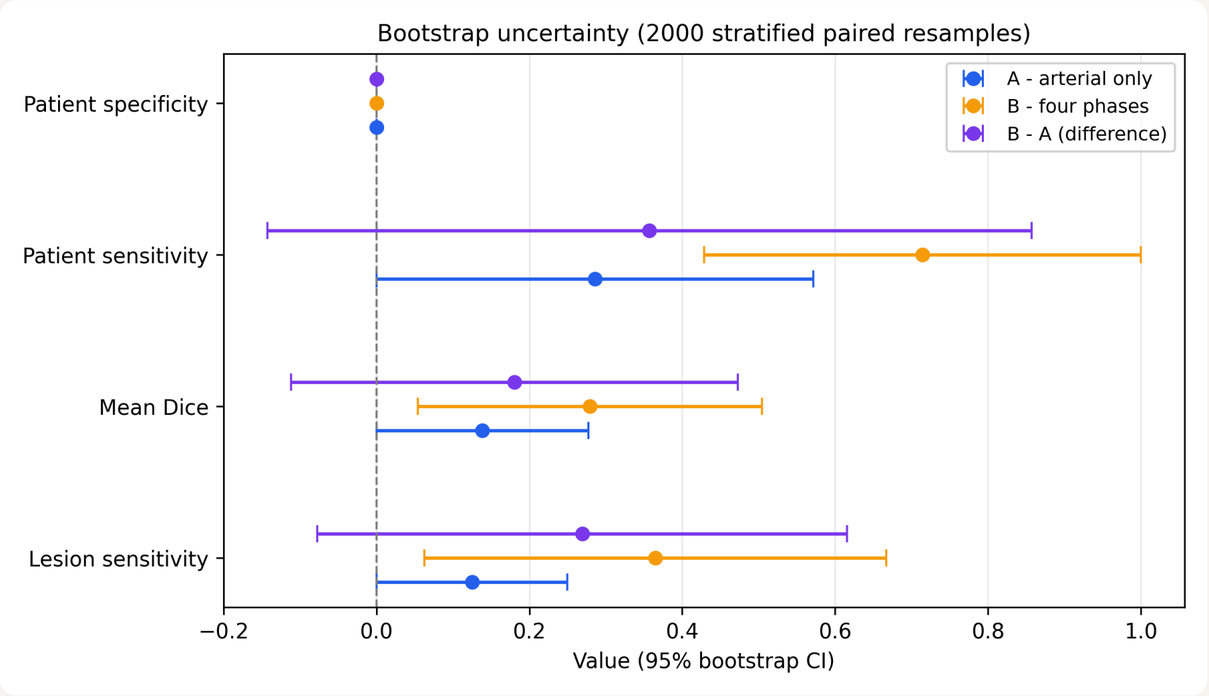
FROC: B lies above A everywhere; at 0.7 FP per patient, sensitivity ≈ **0.33 vs 0.13**.



≤ 20 mm: **0/8 vs 3/8**; > 20 mm: 1/7 vs 2/7 (exploratory only).

RESULTS · UNCERTAINTY

A strong signal, not yet established



B – A lesion sensitivity, 95% CI

[−0.077, 0.615]

B – A mean Dice, 95% CI

[−0.112, 0.473]

Both intervals include zero: with **8 test patients** the study is underpowered. This is a statement about sample size, not about the direction of the effect.

CONCLUSION

The binding constraint is standardized, well-annotated data — not model capacity

Contributions

- Phase-aware, provenance-documented liver MRI dataset
- Leakage-free protocol with frozen thresholds
- Controlled single-variable multiphase test
- Explicit statement of the power limit

Limitations

- 8 test patients, 1 negative; 33:5 imbalance
- Single centre, retrospective, no external test
- Rigid registration only
- Only 24 of 89 lesions biopsy-proven

Future work

- Larger, powered A-vs-B cohort
- Multi-centre, multi-vendor data
- Multi-rater annotation; elastic registration
- False-positive suppression; open artifacts

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